

Chicago Public School Graduation Rate Final Report

Team Project - Georgia Tech M.S. in Analytics Program

I. OVERVIEW OF PROJECT

A. Background

Chicago Public Schools (CPS) serve over 320,000 students, 100,000 of whom are enrolled in high school. CPS faces many challenges: 70.7% of enrolled students are economically disadvantaged [10] and enrollment has declined from a peak of 440,000, leading to the closure of 50 schools in 2013 [9]. Despite these challenges, 4-year CPS high school graduation rates have doubled since 2001, rising from 41.8% to 84.0%. Possible explanations behind the graduation rate trend include a new emphasis on student success during freshmen year [8]; the proliferation of option or alternative schools, which provide a flexible curriculum for students most vulnerable to dropping out; and an increase in students transferring out of CPS, often from the lowest performing districts [7].

B. Problem Overview

This project aims to conduct an explanatory analysis to better understand what factors are most influential to CPS high school graduation rates and to investigate how much of graduation rates can be explained by data released by CPS.

Understanding the factors behind CPS high school graduation rates is important because high school dropouts, on average, have more negative outcomes than high school graduates [3]. According to Levin and Belfield [5], “relative to individuals who complete high school, the average high school dropout costs the economy approximately \$272,000 over his or her lifetime in terms of lower tax contributions, higher reliance on Medicaid and Medicare, higher rates of criminal activity, and higher reliance on welfare”. In addition, regardless of income level, high school dropouts 25 years or older were also reported to be in worse health than high school graduates [6]. While these studies do not prove that dropping out causes these negative outcomes, there are strong associations. As a result, identifying the key factors that influence high school graduation rates is important for creating government policies that improve graduation rates, and, thus, potentially improving the health and economic outcomes of their constituents.

We plan on analyzing these factors through techniques that have a feature importance component, including lasso regression, stepwise regression, and random forests. Furthermore, we intend on using both linear and logistic regression, since graduation rate can be treated as a quasibinomial variable.

C. Initial Hypothesis

Our initial hypothesis is that a school’s graduation rate is most strongly associated with the type of school, its percentage of low income students, attendance, and freshmen on track percentage.

II. OVERVIEW OF DATA

A. Data Source

The first step in our data cleaning process was to download the data. All of our data came from the city of Chicago’s public data portal. We are using multiple datasets, each providing information about different facets of a school’s performance for a given year. If we were to repeat this analysis in future years, we would have to download additional data assets and join them to the existing dataset.

To address this problem, we leveraged the fact that the city of Chicago has made its data available through an open API. This API utilizes the Socrata Open Data API [1]. Using this API, we were able to create a Python script to programmatically scan through the full list of datasets made available by the city of Chicago, find the datasets relating to school performance, and download them to a local directory. This approach makes it significantly easier to scale up our approach or expand to future years as necessary.

B. Data Cleaning

Ultimately, we downloaded 14 data sets: 7 yearly progress reports and 7 yearly school profiles. The progress reports have a variety of metrics for each CPS school including standardized test scores, survey results, discipline rates, and more. The school profiles have demographic information for each school and net new information unrelated to the school’s performance.

CPS published progress reports for school years (SY) 2015-2019 and 2021-2024 and school profiles for SY 2016-2024. Because we wanted to use information from both data sets in our analysis, we only included data from SY 2016-2019 and 2021-2024. The dataset for the 2023-2024 SY is currently just a placeholder, using data from the 2022-2023 SY, so we excluded that data. After filtering the data to only high schools with a value for our dependent variable, 4 year graduation rate, we had 645 observations. This is significantly less than the 1067 high school observations in the combined progress reports and the 1065 high school observations in the combined school profiles.

The downloaded data, particularly the progress reports, had many inconsistencies from year to year. We needed to standardize the names of 50 columns that differed from year to year (e.g. the column with the 4 year graduation rate was named "graduation_4_year_school_1" in some files and "graduation_4_year_school_pct_year_1" in others). Comparing the data from year to year also revealed some new information; for example, CPS stopped reporting average ACT scores after the 2016-2017 SY, moving to the SAT. This coincides with Illinois switching its official standardized test from the ACT to the SAT. [4] Since the ACT and SAT have different scales, with the maximum score being 36 and 1600 respectively, we used imputation to get the SAT scores for 2016-2017 SY.

The school profiles had fewer discrepancies between different years, but had some noticeable discrepancies with progress report data. The school profiles had the same graduation and college enrollment rates for all schools for 2018-2022, so we used the graduation and college enrollment rates from the progress report. We also transformed the demographic raw counts into percentages as percentages are more useful when comparing schools of different sizes.

After combining the duplicate columns via renaming, the combined progress reports had 262 columns and the school profiles had 103 columns. To reduce the number of features, we removed columns where over 50% of observations had missing values. We reviewed the columns under this threshold to make sure none of them were unexpected or likely to contain key information. Lastly, we removed columns that were extraneous (e.g. CPS wide averages) or only populated for middle or elementary schools.

We then removed rows for option/alternative and special education schools. Due to their student population, a few of the key metrics, like freshmen on track percentage and SAT scores, were missing for those schools. Additionally, their graduation rates were extreme outliers relative to other schools. It would be more appropriate to model the factors for these school types on their own. Note that excluding the rows for option/alternative schools does not remove all students who graduated from option/alternative schools from the data; for graduation rates, students should be associated with the school they started at within CPS [7].

Next, all demographic variables pertaining to minorities were dropped during this process except for *student_pct_low_income*. This decision was made out of ethical concerns with the potential later interpretation of any results. *student_pct_low_income* was only kept due to its potential later use as a control variable.

We also decided to remove columns that would result in data leakage. For example, in our initial EDA, overall rating, a composite rating given to each school by CPS, was a very strong predictor of graduation rates. However, upon further investigation, we discovered that graduation rate, our dependent variable, was actually included in the overall rating. We also removed the 5 year graduation rate and the dropout rate for similar reasons.

Lastly, we performed imputation to reduce the amount of

missing values in our data. For each missing value, we first checked if the school had a value for last year or next year and if so, we used that. If there was not an available value for last year or next year, we then checked if the school had a value for two years ago or two years in the future. We performed this process until we reviewed all of the available data. We then dropped columns with more than 10% missing values and imputed the rest of the numeric missing values using the median for the given school year and the non-numeric values using the mode for the given school year.

C. Exploratory Data Analysis

After cleaning, the next step was to perform a preliminary exploration of the data to identify useful correlations and relationships.

The clean dataset contained 49 columns: 48 potential predictors and the single column for the chosen response *graduation_4_year_school_pct_year_2*. The 48 potential predictor columns were then reviewed individually to assess data type. 27 of the columns are categorical variables, whereas 21 were quantitative. 1 variable (SchoolID) and one leftover unnamed variable from the cleaning process were dropped, resulting in 25 usable categorical variables. After identifying this split, we reviewed all predictors again to assess which of these potential variables may be theoretically useful in modeling *graduation_4_year_school_pct_year_2*.

Starting with the quantitative variables, each potential predictor was charted against the response using a scatter plot. Next, we created a correlation matrix to better compare each predictor to the response and to each other. 10 quantitative variables demonstrated a moderate correlation $0.4 < |\text{corr}| < 0.75$ with the response:

- *mobility_rate_pct* : -0.72
- *student_pct_special_ed*: -0.65
- *college_persistence_school_pct_year_2*: +0.64
- *college_enrollment_school_pct_year_2*: +0.60
- *attainment_act_sat_grade_11_pct*: +0.59
- *chronic_truancy_pct*: -0.58
- *sat_grade_11_score_school*: +0.56
- *freshmen_on_track_school_pct_year_2*: +0.43
- *student_attendance_year_2_pct*: +0.43
- *growth_act_sat_grade_11_pct*: +0.41

A subset of the correlation matrix is visualized in Figure 1, displaying the 10 quantitative variables mentioned above.

The remaining variables demonstrated a weak correlation $|\text{corr}| < 0.4$. No quantitative variable demonstrated a strong correlation. These variables may possess outsized explanatory power in later model-fitting.

For the 25 usable categorical variables, a slightly different approach was taken utilizing box-and-whisker plots vs the response. These plots unearthed a few interesting relationships:

- 1) Schools without a dress code saw slightly better graduation rates.
- 2) For schools with recorded "TRUE" values for *refugee_services*, *hard_of_hearing*, and

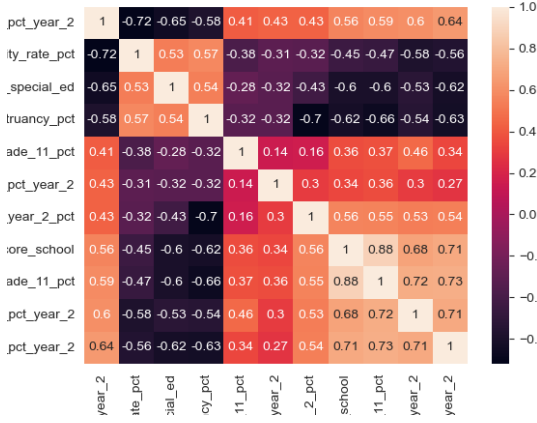


Fig. 1. EDA Quantitative Variable Correlation Matrix

visual_impairments saw notably higher graduation rates than schools with recorded “FALSE” values. The opposite was the case with *after_school_hours_has_variable_times*, where notably higher graduation rates were observed for observations with recorded “FALSE” values. For each of these four variables, this finding is skewed by small sample size; more than 90% of data points fell into one of the two response options.

- 3) For ordinal variables (*student_growth_rating*, *student_attainment_rating*, *culture_climate_rating*, and *creative_school_certification*) as the recorded value increased, so did graduation rates. This is expected as higher scores in these categories likely imply a higher quality school and are correlated with other socioeconomic variables.
- 4) Finally, *school_type* was insightful as rough groups began to materialize by graduation rate over the observed time frame. We can see this relationship in Figure 2. “Selective Enrollment” and “Military” schools possessed the highest median graduation rates (94.50% and 89.05% respectively), followed by “Career Academies”, “Charter”, “Contract, and “Magnet” Schools. “Neighborhood” and “Small” schools fell into the the group with the lowest graduation rates (71.70% and 74.20% respectively).

D. Feature Engineering

After performing data cleaning, we had 27 categorical columns in our dataset. These columns took a variety of forms. Some were simple Booleans. Others, like *long_name*, were unusable for modeling.

There were several survey columns which have multiple potential representations for a regression model. These columns are categorical columns with potential values: VERY WEAK, WEAK, NEUTRAL, STRONG, VERY STRONG, NOT ENOUGH DATA. One potential way to encode these variables would be with a leave-one-out encoding, creating $M = 5$ binary indicator variables. However, this quickly blows

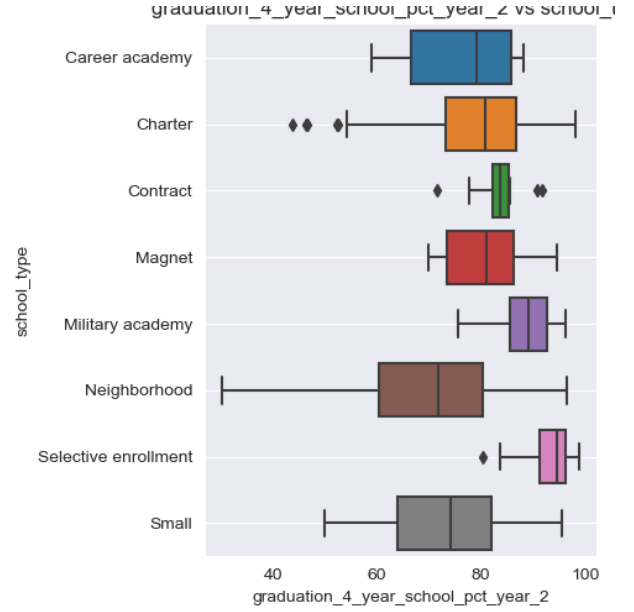


Fig. 2. Box-and-Whiskers Plot of Graduation Rate by School Type

up given we have close to a dozen features in this form. Another option would be to create an ordinal variable where $x = 1$ for “VERY WEAK”, $x = 2$ for “WEAK”, and so on. We would then treat “NOT ENOUGH DATA” as a null value and treat that appropriately. This approach has the advantage of not blowing up the number of columns in our analysis, which decreases the chances of overfitting or losing signal for otherwise significant columns. The disadvantage is that it assumes that there is a linear effect for these variables. For example, if a fitted coefficient b_k corresponds to our ordinal encoding, it assumes that the jump from “VERY WEAK” to “WEAK” is equivalent to the jump from “STRONG” to “VERY STRONG”. To counteract this issue, we could try fitting higher-order terms like x^2 , but doing this increases the number of variables and decreases their interpretability, which undermines the reasons we pursued ordinal encoding in the first place.

Other columns were actually comma-separated lists of values. For example, the *transportation_el* column contains a comma-separated list of Chicago subway lines that a school is near. For example, the value “Green, Red” would denote that a school is near both the green and red lines. To make columns of this form interpretable, we determined the full set of values each list could be populated with. We then created an indicator variable for each of these values, populated with a 1 if a given school has that value in its list of values, and a 0 otherwise. We also added a column denoting the total number of values a given school has for that column. The major columns we were able to process this way were bus routes, subway lines, commuter rail lines, and available language classes. It is possible that treating these columns in this way will introduce too many new columns and lead to overfitting or lost signal. If this turns out to be the case, we can stick with

just the count of total values for a given school. However, features like rail line can be a decent proxy for geography within the city, which could end up being a valuable signal. Another important consideration for this encoding scheme is that by representing an n -dimensional quantity with $n + 1$ terms, we are ruining the linear independence of our terms, which makes regression impossible. To counteract this, we can either use the count term or the one-hot encoding terms, but not both.

One final feature engineering area comes from the *school_hours* and *after_school_hours* columns. These columns are fairly free-form and have inconsistent formatting. A few values could include straightforward ones like "8:00 am - 3:15 pm" or "7:45 - 3:05", or more complicated values like "M, T, Th, F: 8:00am - 3:11pm, W: 8:00am - 3:07pm". There are several schools which have different hours on different days of the week. However, there is generally one dominant set of hours on the majority of days. From that information, we used regular expression matching to isolate: main start time, main end time, hours of class, and a binary indicator denoting whether a school has inconsistent hours throughout the week. These variables could potentially be interesting for investigating the relationship between school start time and academic performance.

E. Data Overview

In the end, we had 595 observations and 45 features to model. The data can be found via this [Dropbox link](#).

III. MODELING & METHODOLOGY

We approached modeling as a two-step process. Our first step was performing feature selection, reducing the large space of available features down to the most meaningful subset. We explored manual feature selection, stepwise regression, lasso, and random forests. Our second step was performing the modeling itself, and we utilized linear and logistic regression for this.

A. Models Used

1) *Linear Regression*: The first model we used was linear regression. In linear regression, we fit an equation of the form:

$$\hat{y} = a_0 + \sum_{i=1}^n a_i x_i$$

where $\hat{y}$ is predicted graduation rate, each x_i is a numeric feature or encoded categorical feature, and each a_i is an associated coefficient, with a_0 being the intercept term.

Graduation rate is a continuous response variable in the range $y \in [0, 1]$, which makes linear regression an appropriate model. We do have to be careful to avoid out-of-sample predictions, however. For example, if x_j is school size and $a_j = 0.003$, then a school with $x_j = 10000$ students could be predicted to have over a 100% graduation rate given the right combination of other factors.

Linear regression is an especially useful tool for us because of the straightforwardness of interpretation. For a given coefficient a_j , we know that a one-unit increase in x_j on average corresponds to an a_j increase in graduation rate. Linear regression is also versatile enough to model a variety of relationships through interaction terms and variable transformations.

While linear regression on its own is a versatile tool, we used a few different variations of it to choose an appropriate feature set and avoid overfitting. These tools included stepwise regression and lasso regression.

2) *Logistic Regression*: Logistic regression is a modified version of linear regression. While linear regression assumes a linear relationship between the predictors and the response, logistic regression is instead based on the assumption that the response variable is drawn from a binomial distribution. We then get an equation of the form:

$$\text{logit}(p) = \log \frac{p}{1-p} = a_0 + \sum_{i=1}^n a_i x_i$$

Here, each a_j and x_j is the same as in the definition of linear regression. The value p is the predicted probability of our output being equal to 1.

Technically, our dependent variable is not drawn from a binomial distribution. However, it is very close. Graduation rate can itself be thought as being a parameter of a binomial distribution, where that distribution models the probability of a given high school student graduating. Therefore, it can make sense for us to model graduation rate using logistic regression.

Within R, we utilized the GLM function to implement logistic regression. Within this function, we utilized the quasibinomial model family. This family captures the idea of not being true logistic regression, but rather modeling a univariate probability distribution.

One advantage logistic regression has in the context of this problem is variable interpretation. For a given variable x_j , a 1-unit increase in that value is associated with a $100a_j\%$ increase in the odds of a student graduating from a given school. This also solves the boundedness issue we had with linear regression since predictions are constrained to the same $[0, 1]$ range as the response variable.

B. Feature Selection Methods

1) *Stepwise Regression*: Stepwise regression is an iterative approach to feature selection in linear regression. It comes in a few different forms:

- **Forward Selection**: Here, we start with an empty model $\hat{y} = a_0$ and iteratively add features. At each step, we add the feature which produces the biggest gain in performance according to some selection criterion. For our models, we used the Akaike Information Criterion (AIC). We continue this iteration until there are no more features whose addition produces a better model. As a final step, we prune insignificant features at the end.
- **Backward Elimination**: Here, we start with all features present and iteratively remove features. At each step, we remove the worst-fitting feature according to AIC. We

continue this iteration until the fitted model degrades past a certain threshold.

- **Bi-Directional Selection:** This is a combination of both forward selection and backward elimination. Similar to forward selection, we start with an empty model, but at each step we now consider adding or removing features. Using this method, a feature could be added to the model in an early step but then removed in a later step if it leads to the biggest improvement in the AIC of the model. We continue this iteration until there are no features to add or remove that would improve the model past a certain threshold.

These approaches all have the advantage that at the end, you are left with a model based on a set of mutually significant features. It is also a fairly straightforward model, with few hyperparameters to tune beyond the selection criterion and certain cutoffs for model performance. This makes it an ideal baseline to compare more complicated approaches against, as well as a competent model in its own right. We felt that stepwise regression was a good approach to test due to it being an iterative process that can be useful when dealing with many predictor variables.

Ultimately, these approaches all resulted in the same 21 significant features.

2) *Lasso*: Lasso regression is a modification to traditional linear regression to incorporate feature regularization and selection. Lasso regression offers several advantages over traditional linear regression, including automated feature selection, handling of multicollinearity, and prevention of overfitting. By shrinking coefficients towards zero, lasso promotes model interpretability and manages high-dimensional datasets effectively.

Lasso achieves these effects by adding an l_1 regularization term to the loss function. That is, we add $\min \lambda \sum_{i=1}^n |a_i|$ to the objective function. The hyperparameter λ controls the strength of the penalty, with higher values assigning a harsher penalty to large coefficients.

Initially, we attempted to build a lasso regression model using all features, but this did not yield meaningful results. Subsequently, we selected 10 features that we believed could be significant predictors based on theoretical relevance.

- student_attendance_year_2_pct
- sat_grade_11_score_school
- mobility_rate_pct
- school_type
- student_growth_rating
- student_attainment_rating
- culture_climate_rating
- school_survey_student_response_rate_pct
- school_survey_teacher_response_rate_pct
- school_survey_involved_families

With these 10 features we leveraged R functionality that found the optimal lambda based on the lowest error from different cross validated models. Lastly, we modeled the features amongst with non-zero coefficients in a linear regression without regularization to find the R^2 .

C. Random Forest

The final model we utilized for our analysis is a random forest. Random forests are based on the concept of decision trees. Rather than taking into account each feature at once the way regression does, decision trees instead operate by making a series of splits in the data. They can optimize a number of metrics, but these metrics generally tend to involve splitting data into groups to minimize the intra-group entropy. The end result is something like a flow chart, where we start at the root node and descend downwards, at each step looking at the split variable and moving to the appropriate node.

One major weakness of decision trees is that they are somewhat easy to overfit. Random forests are a powerful extension of decision trees which overcome this limitation by constructing many (we used an $n = 50$) smaller decision trees with slightly different splits. Then, to classify a single data point, we get predictions from each individual decision tree and average the results together.

Random forest models have several advantages over linear models: they can handle non-linear relationships and high-dimensional data. They also are not constrained by assumptions about the underlying data distribution in the way that regression-based models are. Random forest models provide feature importance rankings by looking at which features are most commonly at the top of the tree and thus contribute to the splitting of the most samples. Furthermore, the fact that random forests are an ensemble method (or a combination of several models) often leads to enhanced predictive performance.

We applied Random Forest variable selection to all numerical variables in our dataset and identified the top 15 features. We then modeled these top 15 features in a linear regression to find the R^2 .

IV. RESULTS

One important thing to note is that we reran our models between the progress report and final report. The progress report data did not include imputed values, and we added a handful more features in the interim. For the progress report, we excluded rows that had missing values, which resulted in a smaller and more biased data set. Our results have slightly worse performance (as measured by R^2), but better reflect all high schools in the CPS system.

A. Key Variables

Each of the aforementioned models had a different approach to finding the key variables that most impacted graduation rates. The results can be found in Table I on page 6. If the model has a numeric value, the model included feature importance, with 1 indicating the most important feature; otherwise, an "X" indicates that the variable was chosen by the model.

There was only one variable chosen by all four models: mobility rate percentage. The student mobility rate is the percentage of students who experienced at least one transfer in or out of a school during the school year, excluding graduation.

TABLE I
KEY VARIABLES

Feature	Stepwise	Lasso	Logistic	Random Forest
attainment_act_sat_grade_11_pct				6
average_length_suspension_year_2_pct	X			
beginning_year	X			14
chronic_truancy_pct				7
classroom_languages			X	
college_enrollment_school_pct_year_2				10
college_persistence_school_pct_year_2	X			4
culture_climate_rating	X	X	X	
freshmen_on_track_school_pct_year_2	X		X	3
growth_act_sat_grade_11_pct			X	12
mobility_rate_pct	X	X	X	1
network	X		X	
refugee_services	X		X	
sat_grade_11_score_school				5
school_hours_num_hours	X			
school_survey_ambitious_instruction		X	X	
school_survey_collaborative_teachers		X		
school_survey_involved_families	X	X	X	
school_survey_student_response_rate_pct	X			
school_survey_supportive_environment		X		
school_type	X		X	
student_attainment_rating	X			9
student_attendance_year_2_pct	X	X		15
student_count_total			X	8
student_growth_rating	X		X	
student_pct_english_learners	X			
student_pct_low_income	X			
student_pct_special_ed	X		X	2
suspensions_per_100_students_year_2_pct				13
transportation_el_num_distinct	X			
transportation_metra_num_distinct	X		X	
visual_impairments	X			

It can be triggered by a voluntary event, like a family move, or involuntary events, like homelessness or changing schools due to bullying or expulsion. Transfers often result in academic and social challenges and are associated with lower graduation rates and delayed graduation [11]. Being chosen by all four models, our findings on student mobility are consistent with the other research highlighting its importance.

Culture climate rating, freshmen on track percentage, school survey involved families, student attendance, and student percent special education were selected by three out of the four models. Culture climate rating is an aggregate score from the different school surveys administered to both students and teachers, composed of the 5 school survey features: ambitious instruction, collaborative teachers, involved families, supportive environment, and effective leaders. Involved families is a survey metric measured by teachers. Freshmen on track percentage was one of the features we expected to be significant due to research performed by UChicago and cited by CPS officials [8]. We also expected student attendance to be selected by most models. Percent of special education students is also consistent with our findings from data cleaning and EDA, where we found special education schools had significantly lower graduation rates.

The other two variables we hypothesized would be important, school type and low income student percentage, were selected by two and one models, respectively. This indicates that the variables are somewhat important, but not as impor-

TABLE II
PERFORMANCE OF MODELS

Model	Adj R^2
Stepwise Linear Regression	0.75
Linear Regression (Random Forest Variables)	0.71
Lasso Linear Regression	0.63
Logistic Regression	0.72

tant as the other two variables, freshmen on track rate and attendance.

Interestingly, features related to SAT scores were found to be important by the random forest model, but not by the linear models, with the exception of SAT growth percentage by logistic regression. This suggests SAT scores may have a non linear relationship with graduation rate.

B. Model Performance

The results can be found in table II on page 6.

All three of the stepwise regression methods resulted in the same variables selected, and therefore the same model. Each approach selected 21 variables, leading to an $R^2 = 0.76$ and adjusted $R^2 = 0.75$, suggesting that the model explains approximately three quarters of the variability in the graduation rates of CPS schools. The p-value for the overall model was $2.2 \cdot 10^{-16}$, meaning that the overall model was statistically significant.

To test the performance of variable selection from the random forest model, we took the 15 most important variables and ran a linear regression to find the adjusted R^2 ; this enabled us to perform an apples-to-apples comparison with the other linear regression models. The linear model fit to the 15 most important variables had an adjusted $R^2 = 0.71$, meaning we lost some performance compared with the stepwise models.

Linear regression with the features chosen by the optimal lasso regression had a relatively low adjusted $R^2 = 0.63$. This may be related to the manual, initial filtering of variables we tested for the lasso regression or the cross validation performed when choosing the optimal λ value for the lasso regression; the latter prioritized features that were less prone to overfitting.

Our logistic regression model was trained on a manually selected list of 18 features. It had an adjusted $R^2 = 0.72$, which gave it comparable performance to the random forest-derived model. We had an alternate logistic regression model trained on the full dataset with an adjusted $R^2 = 0.79$. However, this model was trained on over 200 features when categorical features are taken into account, which means that the model was likely massively overfitting.

C. Model Usefulness

1) *Stepwise Regression*: The stepwise regression methods demonstrate some usefulness. More than half of the variables from the dataset were discarded (only 21 of the 45 potential variables were selected), meaning that the discarded variables are likely not useful in predicting graduation rates of CPS schools. We did not get any feature importance using this methodology, so we had to rely on other models to determine which features were the most important amongst the 21 significant features.

Of the 21 variables selected, several were variables we expected to have explanatory power before we began our analysis, including:

- school_type
- student_attendance_year_2_pct
- student_pct_low_income
- freshmen_on_track_school_pct_year_2

Additionally, there were some selected variables that would be interesting to look further into, such as *beginning_year*. This variable had a p-value of approximately 0 in the stepwise regression model, indicating that it is highly correlated with graduation rates of CPS schools. It would be interesting to look into what factors might have been different in each year of our dataset to understand why graduation rates would be different across the years.

2) *Random Forest*: The random forest model was useful as it clearly provided feature importance and considered non-linear relationships between features. Notably, random forest highlighted the importance of the freshmen on track percentage, which was #3 in importance. However, two features we expected to be significant, school type and percent of low income students, were not in the top 15 important features.

3) *Lasso*: The lasso linear regression had more limited usefulness since it did not perform feature selection when modeled on all of our features. As a result, we performed manual feature selection in order to prepare lasso for feature selection. Interestingly, all three features eliminated by lasso regression were found to be relevant by other models. In addition, we expected school type, which was eliminated, to be significant at the outset of the project.

4) *Logistic Regression*: Due to the hand-tailored features we used in defining our logistic regression, a few interesting features came out of this analysis. One was the ISP network of schools being a strong positive predictor of graduation rate. Upon further research, we learned that ISP stands for "Independent School Principal". It is a program where exceptional principals are given more autonomy to set budgets and dictate other administrative tasks [2]. It is somewhat difficult to disentangle the causality here, since the program is only offered to exemplary school principals. Are these strong graduation rates caused by the skill of the principal or the extra leeway offered by this program?

Independent of our full logistic regression model, we also performed a few regressions on small subsets of the overall feature space. In doing so, we made some interesting observations. One was that just by modeling on what bus routes a school has access to, we could get an $R^2 = 0.603$. This can be seen as a proxy for neighborhood or geography. Since bus routes in Chicago tend to follow along a single street or within a neighborhood, bus routes are fairly robust indicators of location within a city. Overall, this shows the heavy degree to which location plays a factor in graduation rate. It also shows that if the City of Chicago wants to raise graduation rates in some of the worse-off schools, they may need to pursue more structural changes in the neighborhoods those schools serve.

Another observation from this analysis came from building a model on student demographics. We did not want to use this data in our broader modeling to avoid biasing our results. However, when looked at in isolation, we found that a school's racial profile did not have a significant impact on graduation rate. The only demographic factors which seemed to have an impact were special ed and English learners. These are both groups which need extra support to succeed in high school, and they could both be potential targets for intervention to improve outcomes.

D. Hypothesis Evaluation

Our initial hypothesis was that a school's graduation rate is most strongly associated with the type of school, its low income students percentage, attendance, and freshmen on track percentage.

Our hypothesis was close to our findings; student attendance and freshmen on track percentage were each selected in three out of our four models, school type two out of four models, and low income students percentage one out of four models. The impact of school type may have been subdued due to our exclusion of alternative and special education schools during

data cleaning. However, low income students percentage was not as important as hypothesized; however, this feature may be correlated with other selected factors like mobility rate.

Modeling brought new features to light, particularly mobility rate percentage, survey data, and percentage of special education students.

V. UNEXPECTED CHALLENGES

Data cleaning was more difficult and time consuming than we had anticipated. Formatting did not change significantly from year to year, but there were many small changes that added up and were difficult to catch. These changes are reviewed in the data section, but some trends include slightly changing column names and changing ordinal variable values.

In addition, missing data was pervasive, and aside from the relationship to school type (alternative and special education schools), random. Due to the amount of effort it took to clean the data, we performed initial modeling with non-imputed data, performed imputation, and then ran our final set of models.

Lastly, there was no centralized, detailed document that defined each variable and went into their methodology; we had to aggregate information from different CPS documents and from outside sources to understand the data definitions.

VI. FUTURE WORK

In future iterations of this analysis, there are four more topics worthy of investigation.

A. Goodness-of-Fit Testing

One area we would like to work more on is model validation. In order to ensure that our results are statistically sound, we could have benefited from performing a residual analysis. Doing so would allow us to more rigorously interrogate our assumptions. If we found that these assumptions are not valid, we could potentially simplify our approach or pivot to different models.

B. More Years of Data

We worked with data from the 2016-2024 time period for our analysis. Compared with the overall history of the Chicago Public School system, this is a comparatively brief timeline. If we could work with the City of Chicago to obtain records from previous years, we could expand our analysis in a number of ways:

- **Event Detection:** Using change detection techniques like CUSUM, we could search for sudden events in the graduation rate and its influencing factors.
- **Trend Decomposition:** Using trend decomposition techniques like exponential smoothing, we could understand gradual trends in graduation rates over time.
- **Robustness Checks:** Using more years worth of data would enable us to ensure that our techniques are properly generalizable and give us more confidence in our resulting assertions.

C. Outside Data

There is a lot that can be learned from data outside of the CPS data. We have already started to approximate some of this data from things like proximity to different rail and bus lines, but we could potentially go much farther.

- **Neighborhood Data:** How do factors like average property value, neighborhood walkability, access to amenities, etc. impact graduation rates?
- **Crime Rates:** In the City of Chicago, there are a few neighborhoods which are subject to elevated gang activity and other violent crime. It makes sense that living in these kinds of conditions would have a deleterious impact on a student's ability to even attend school, let alone excel and graduate.
- **Macroeconomic Trends:** If we had data over a long enough time scale, how do macroeconomic trends impact graduation rates? For example, what did graduation rates look like during the 2008 recession?

D. Causal Inference

From a methodological perspective, one thing which would be fascinating to look into would be the causal effect that these various features have on graduation rates. If a policymaker was trying to leverage our results, there is some ambiguity in what can be done to improve graduation rates given our findings. For example, we found that the Independent School Principal (ISP) program is predictive of a higher graduation rate. However, what is the actual causal lever at play here? Is it the skill of the principal? The increased freedom they are given in the program? A combination of both? Some other factor? It is important for policymakers to know how the various levers actually available to them work.

There are a few different ways we could perform causal inference. Randomized controlled trials would likely be infeasible due to the amount of resources they require and the ethics of performing certain types of interventions. Natural experiments would be ideal, but they are also somewhat circumstance-dependent.

VII. CONCLUSION

In summary, the 4-year CPS high school graduation rates have doubled over the last twenty years [8]. This project aims to conduct an explanatory analysis of the drivers of high school graduation rates, utilizing data provided by CPS.

Our initial hypothesis was that a high school graduation rate would be mostly strongly associated with school type, percentage of low income students, attendance, and freshman on track percentage.

The evaluation of this hypothesis was tackled in two parts: 1) Feature Selection and 2) Model Fitting. For the former, manual feature selection, stepwise regression, lasso, and random forests were employed to identify the most meaningful subset of predictors. These predictors were then processed by linear and logistic regression. Ultimately, this resulted in four final models: 1) Stepwise Linear Regression, 2) Linear Regression (Random Forest Variables), 3) Lasso Linear Regression,

and 4) Logistic Regression. The Adjusted R^2 of these models ranged from 0.63-0.75, implying the majority of the variation in high school graduation rates is explained by our models. The features chosen across the four models were then examined to determine whether our initial hypothesis was supported.

Returning to the hypothesis, student attendance and freshman on track percentage were selected in three of the four models. School type was selected in two of four models. Percentage of low income students was selected by only one model. As this variable was kept (vs other demographic variables) mainly for controlling purposes, the lack of significance is not necessarily surprising. Interestingly, mobility rate percentage was the only variable selected by all four models, implying higher importance than originally expected.

In terms of causal inference and policy recommendations for further improvement of CPS high school graduation rates, we admit practical challenges in converting our results to actionable policies. While we believe our results are valuable in identifying the key attributes explaining historical high school graduation rates, crafting policy to affect impact high-importance variables (student attendance, freshman on track percentage, mobility rate percentage, etc.) may prove challenging. The drivers of these variables are multi-faceted and influenced by larger socio-cultural, economic, geographical, and demographic trends – some unique to the history of the city of Chicago while others are derivative of broader national trends.

Despite these challenges, we hope that our results provide a starting place for Chicago policymakers to better understand the drivers of high school graduation rates in hopes of continued efforts to improve educational outcomes for the benefit of the city of Chicago and the health and economic outcomes of her citizens.

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